

Passing Loebner's Turing Test: A Case of Conflicting Discourse Functions¹

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Abstract. This paper argues that the Turing test is based on a fixed and de-contextualized view of communicative competence. According to this view, a machine that passes the test will be able to communicate effectively in a variety of other situations. But the de-contextualized view ignores the relationship between language and social context, or, to put it another way, the extent to which speakers respond dynamically to variations in discourse function, formality level, social distance/solidarity among participants, and participants' relative degrees of power and status (Holmes, 1992). In the case of the Loebner Contest, a present day version of the Turing test, the social context of interaction can be interpreted in conflicting ways. For example, Loebner discourse is defined 1) as a friendly, casual conversation between two strangers of equal power, and 2) as a one-way transaction in which judges control the conversational floor in an attempt to expose contestants that are not human. This conflict in discourse function is irrelevant so long as the goal of the contest is to ensure that only thinking, human entities pass the test. But if the function of Loebner discourse is to encourage the production of software that can pass for human on the level of conversational ability, then the contest designers need to resolve this ambiguity in discourse function, and thus also come to terms with the kind of competence they are trying to measure.

Key words: communicative competence, cooperative principle, discourse function, Grice, linguistic politeness, Loebner Contest, Turing test

This line of teleology for the Loebner prize, that it serves not as a test of the abilities of the computers but of the psychologies of the various participants, has often been proposed informally.

Stuart M. Shieber²

After the main event, all the guys assured me with knowing smiles that I was "very convincing" – which made me wonder if the contest wasn't just an elaborate joke propagated by the AI community upon a few chosen humans.

Tracy Quan³

1. Introduction

Fifty years ago, mathematician Alan M. Turing (1950, p. 433) proposed a test – he called it the "imitation game" – that was designed to simplify and clarify the philosophical and practical problems of defining and measuring machine intelligence. According to the rules of the imitation game, a human interrogator attempts to distinguish two players, a human and a machine, on the basis of their teletyped responses to his/her questions.⁴ A machine passes the test (i.e., it is deemed a "thinking" entity) if the interrogator can not tell the difference between the players



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after “five minutes of questioning” (1950, p. 442). Due in part to its severity as a scientific test (Moor 1992, p. 1625), the imitation game (afterwards known as the Turing test) has proven remarkably durable. Because the interrogator is free to ask any number of silly or irrelevant questions, the test appears to offer a very efficient method for separating nonhumans from humans (Dennett, 1985). As Turing says (1950, p. 435), “The question and answer method seems to be suitable for introducing almost any one of the fields of human endeavor that we wish to include.”

But while the test does not restrict what the interrogator can say or ask, it does appear to restrict how s/he can say it and how players can respond. The interrogation model of discourse creates fairly rigid, clearly distinct, roles for participants. The interrogator controls the flow and topic of conversation; the subjects of the interrogation are powerless, except to the extent that their responses provide the information the interrogator is looking for. The interrogator, of course, is free to ask a variety of different questions. But these questions (including how the interrogator asks them) will necessarily be shaped by the function of discourse, which in the case of an interrogation is to force an unwilling participant to reveal, perhaps inadvertently, some piece of important and concealed information.

Turing test advocates assume that any machine that can pass a serious interrogation will be more than fit to interact with humans in an indeterminate number of situations. In other words, they assume that communicative competence is fixed and transferable: a machine that passes a Turing test will also be capable of passing for a thinking entity in other settings. But a machine that can pass a serious interrogation will not necessarily make a good conversationalist. This machine may not be equipped with the ability to direct the flow of conversation, especially if it has been programmed to play the role of subject in an interrogation. It may not be able to appeal to politeness conventions where appropriate or adjust to changing power relationships. It may not be able to use humor – or be indirect, change the subject deftly, deploy face-saving strategies, or do any of a number of other things – in order to deflate potentially volatile situations. What is needed, in other words, is an alternative to the fixed and de-contextualized view, one that explores the ways in which *context* shapes what it means to be a competent speaker/writer. Because a contextual view treats competence in terms of a speaker’s dynamic response to variations in discourse function, formality level, social distance/solidarity among participants, and participants’ relative degrees of power and status (Holmes, 1992), it calls into question whether a program designed solely to survive a Turing test will also survive in situations where the avoidance of conflict and the maintenance of social harmony are expected (Kasper, 1990).⁵

This paper argues that test designers need to attend more carefully and explicitly to the ways in which discourse is shaped along the dimensions of communicative competence. For example, in the case of the Loebner Contest (Loebner, 1999), a present day variant of the Turing test, discourse function can be interpreted in two conflicting ways (i.e., discourse-as-interaction and discourse-as-transaction).

This suggests that contest designers have either failed to come to terms with the interrelationship of language and social context, or assume that the ability to communicate transcends any specific context. It is my hope that the analysis of Loebner Contest transcripts that follows will demonstrate the danger in thinking of language in isolation from social context.

2. The First Turing Test

Billed as the "First Turing Test" (Loebner, 1999), The Loebner Prize for Artificial Intelligence is an annual contest sponsored by New York City businessman Hugh Loebner, who has promised \$100,000 and a gold medal to the first program to pass an unrestricted Turing test. Each year, the prize committee awards a nominal amount (originally \$1500, currently \$2000) and bronze medal to the "most human" computer program entered in the contest that year. The contest has been held every year since 1991.

The Loebner planning committee made two important revisions to the structure of Turing's original game. Turing's game is based on a tripartite structure, in which each player (human and software program) can view and respond to the utterances of the other two parties. The Loebner competition, on the other hand, is organized around a set of dyadic interactions between judges and contestants. The judges are informed that at least a certain number of contestants may be human "confederates."⁶ Each judge interacts with one of the contestants for a specified amount of time (ten minutes in 1999), then moves to another terminal and begins another conversation with a new contestant. This process is repeated until every judge has interacted with every contestant. It is each judge's job to rank all the contestants in order from "least human" to "most human." The judges are also asked to rank each contestant on scales of "responsiveness" and "humanness," although these rankings are used only in the event of a tie. Each contestant's median score is used to compute an overall ranking, and the program with the highest ranking is the prizewinner. Should a program fool 30% of the judges⁷ into mistaking it for human (it has not happened yet), it will be awarded an intermediate prize of \$25,000 and a silver medal. (The human confederates always outrank the software programs.) During the second phase of the contest, software programs will not only be required to pass a text-only version of the game but also to "respond 'intelligently' to audio visual input in the manner that a human would" (Loebner, 1994). The first program to pass this second, audiovisual test (presumably at Turing's 30% level) will be awarded the \$100,000 grand prize.

The second revision to the structure of Turing's game came as the result of the planning committee's realization that, "given the current state of the art, there was no chance that Turing's test, as originally defined, had the slightest chance of being passed by a computer program" (Shieber, 1994, p. 70). The committee thus decided to limit the range of what each entrant would be responsible for discussing. In the case of the programs, designers defined topics in advance (e.g.,

pets, hockey, sex, whimsical conversation, and burgundy wines). In the case of the human confederates, the prize committee chose topics that intersected with each confederate's area of interest or expertise. In one well-known example from the first Loebner competition (held in 1991), confederate Cynthia Clay, whose selected topic was Shakespeare's plays, was mistaken for a program three times (Shieber, 1994; Hutchens, 1996). One judge said, "[no] human would have that amount of knowledge about Shakespeare" (Stipp, 1991, quoted in Shieber, 1994, p. 72). Judges were required to stick to each entrant's pre-defined topic. They were also told to refrain from "trickery or guile" (Shieber, 1994, p. 71): obscure or unexpected questions and gibberish designed to expose nonhumans would not be allowed. Finally, the rules stated that "Judges should respond naturally, as they would in a conversation with another person" (Shieber, 1994, p. 71). Referees tried to ensure that judges and confederates – a new batch was brought in each year – acted in accordance with these rules. The topic restriction spanned four contests (1991–1994); the contest has been unrestricted since 1995.

Loebner judges have made fine interrogators; Turing would no doubt be proud. Judges typically presume prior to interaction that all contestants are nonhuman until proven otherwise. Judges see their task as one of forcing contestants to reveal (or to make obvious) the truth of their hidden identity. Many rely on what I call pre-selected "outing" questions, the majority of which require from contestants a broad knowledge of facts about history and culture. Outing questions are similar to what Dennett (1985, p. 126) calls "quick probes," in other words questions or statements designed to "test for a wider competence." Dennett (1985, p. 138) advocates quick probes as a solution to the problem of "putting too much faith in computer systems, especially user-friendly systems that present themselves in a very anthropomorphic manner." Loebner judges expect answers to these outing questions or quick probes to be concise, direct, and unmitigated; contestants who/that try to change the subject – let alone direct the flow of conversation – are treated to a quick and merciless onslaught of discursive harassment.

Put another way, many judges resemble vigilantes who police the boundary between the human and the nonhuman. While the vigilante approach can "teach all users of computers, especially users of expert systems, how to probe their systems before they rely on them" (Dennett, 1985, p. 138), it puts a serious strain on conversation. Because judges do not take contestants' humanness for granted, they are unable and unwilling to trust them enough to interact in ways we would expect from two people who do not know each other very well or at all. In casual conversation, we typically expect each participant to engage in a fair amount of interpretive work in order to make sense of the direction and meaning of the developing conversation. Many Loebner judges, however, view interaction with contestants in terms of a one-way transaction in which they alone control the conversational floor. Because judges expect responses from contestants to be direct and complete, they are unwilling to work with anything that might be grammatically or conceptually incomplete (Collins, 1997). In the rest of this paper, I would like

Table I. Loebner Discourse as Casual and Transactional

Casual Discourse	Transactional Discourse
• Humanness: each participant assumes the other is human <i>prior</i> to interaction.	• Nonhumanness: interrogator doubts the other's self-presentation as a human (Note that this does not hold for face-to-face interaction between two humans.)
• Politeness strategies	• Impoliteness strategies, rudeness
• Politeness Principle (responses will take into account the level of imposition, and the other's social distance and relative power)	• Cooperative Principle (responses will be length-appropriate, truthful, relevant, and perspicuous)
• Interactional or casual	• Transactional or fact-based
• Two-way: both participants share the conversational floor.	• One-way: interrogator controls the conversational floor; players are powerless.
• Programmers & Researchers (build "personalities" for interaction)	• Judges (expect compliance with maxims of Cooperative Principle, though they themselves are not bound to them)
• Loebner Contest	• Loebner Contest

to draw on selected research in sociolinguistics to build a model of casual conversation between human strangers, and compare it with the model of communicative interaction presented in the Loebner competition rules and in interactions between judges and contestants. The set of distinctions I have been sketching here loosely are organized in Table I. Because the function of Loebner discourse is ambiguous, I have placed it on both sides of the conversational divide. This is not cause for concern as long as the goal of the contest is to ensure that only thinking entities pass the test. (The judges are doing a great job of that now.) But if the goal of the contest is to encourage the production of software that can pass for human on the level of conversational ability, then the contest designers need to resolve this ambiguity in discourse function, and thus also come to terms with the kind of competence they are trying to measure.

Table II. Complete list of topics during the restricted Loebner era (1991–1994)

Program Topics	Human Confederate Topics
Bad Marriage	Abortion
Burgundy Wine	American History
Cats versus Dogs	Baseball
Classic Star Trek	Boston Red Sox
Dry Martinis	Classical Music
Environmental Issues	Cooking
Hockey	Cryonics
Liberal or Conservative	Custom Cars
Men vs. Women	Dream Analysis
The O.J. Simpson Trial	Health
Pets	Martial Arts
Problems in Romantic Relationships	Newspaper Comics
Relations with your real or imaginary wife	The Rolling Stones
Second Grade School Topics	Shakespeare's Plays
Sex Education	Women's Clothing
Small Talk	
Whimsical Conversation	

3. Playing It Safe

In designing the first contest, the Loebner committee preferred to play it safe. According to Robert Epstein (1992, p. 84), who established the contest on behalf of Loebner, “[the planning committee] took great pains to protect the computers. We felt that in the early years of the contest, such protection would be essential.” This protection, as described above, took the form of a restriction in topic. In choosing topics, program designers and the Loebner committee generally shied away from topics that might sanction discursive harassment or suggest that programs were not to be protected after all. See Table II for the complete list of topics during the restricted Loebner era (1991–1994) (sources: Mauldin, 1994; Platt, 1995). Of course, any topic can fuel aggressive behavior. Most of the topics on this list, however, are apparently intended to be on the innocuous side. They are topics, as Shieber (1994, p. 72) suggests in reference to the 1991 competition, “appropriate for a cocktail party venue.” Most fall into the category of leisure and recreation. Nearly all are topics that permit participants to offer their opinions without also having to worry whether they are offending their listeners. In other words, it is unlikely that your conversational partner is going to feel threatened if, say, your favorite episodes of *Star Trek* do not coincide, or your classical music tastes differ, or one of you likes

dogs better than cats. Because the majority of these topics are concerned with taste (which is assumed to vary widely among individuals), differences of opinion are expected but not expected to cause party guests to go to blows.

Shieber (1994, p. 71) reports that, during the first Loebner Competition, judges and confederates “performed admirably”: they stuck to assigned topics and refrained from so-called “trickery or guile.” But transcripts of interactions from restricted Loebner contests clearly reveal tensions among participants. Judges did not blindly follow the instruction to “respond naturally, as they would in a conversation with another person,” simply because the very thing assumed by that instruction – namely, the personhood or humanness of each contestant – was in serious doubt. In the following example, taken from the 1993 competition, a judge and a program are supposed to be discussing the topic “pets”:⁸

FROM 1993, DESIGNER M. MAULDIN, RANKED 3RD OUT OF 3 PROGRAMS, JUDGE NUMBER NOT AVAILABLE

Judge Have you ever kept a penthouse pet?
Program Do you have any pets?
Judge I think you're avoiding my question.
Program Do you think that animals can think?
Judge Do you think it ethical for one species to enslave another
Program What kinds of pets did you have as a child?
Judge I'm asking the questions. What is a pet if it is not a
 slave of species-ism.
Program Oh, that's so sad. I think pets enrich a childhood
 immensely.
Judge How can we keep some animals as pets, yet eat others
Program Guinea pigs are about as smart as cumcubmers.
Judge Would you eat a guinea pig
Program I spent several summers on my Uncle's farm. I really
 enjoyed the animals a lot.

In this example, the judge manages at once to assume a hostile persona and stay on topic, which suggests that the restrictions did not always function as intended.

In 1995, the topic restriction was lifted. But the contest rules have continued to maintain a set of implied restrictions. For example, rule five attempts to elicit polite behavior from judges and contestants by instilling in them (or their designers) a fear of reprisal (Loebner Prize Rules for 2000):

5. It is the task of the computer entries to respond to the communications of the judges in such a manner as to imitate the responses of a human. No constraints will be placed on the judges', confederates' or contestants' conversations. How-

ever, the competition is widely publicized, the transcripts will be published, and all participants must assume responsibility [sic] for their own contributions. As in any normal conversation [sic], both participants have a role in determining the direction of the conversation and are free to decide what [sic] topics they do or do not wish to discuss. Entries must be prepared to communicate for an indefinite period of time.

That the contest is “widely publicized” is offered as a warning to those participants who might be tempted by the possibility of unconstrained communication. The judges, program designers, and confederates are encouraged here to internalize a set of rules that were previously explicit: the embarrassment and humiliation of a public reprimand – one with an indefinite life span – is meant to keep participants in line.

In addition to instilling participants with a fear of (public) reprisal, the post-restriction Loebner rules also attempt to enforce non-threatening discourse by masking the game’s function as a task-oriented interrogation. For example, the rules (see rule five again) define communication as a reciprocal and “normal” affair between two participants of equal power. In addition, two supporting documents, “Instructions for Confederates” (1999) and “Instructions for Judges” (1999), encourage the humans to “Enjoy yourself! It should be just like striking up a conversation with the person sitting next to you on a train/plane trip.” The train/plane analogy suggests that the function of Loebner discourse is to interact: the primary goal of such discourse is the “establishment and maintenance of social relationships” (Kasper, 1990, p. 205).

4. The Politeness of Strangers

In “normal” conversation, we expect non-intimates to adhere to politeness rules, with each party striving to avoid conflict and to maintain social harmony (Kasper, 1994; Culpeper, 1996). Much of the interest in linguistic politeness over the last two decades stems from the work of Brown and Levinson (1987 [1979]), who argue, building on Goffman’s notion of “face” (see Goffman, 1955, 1967), that speakers typically avoid speech acts that threaten their own face or the faces of other participants. Should a face-threatening act (FTA) prove unavoidable, Brown and Levinson suggest that a speaker will typically try to disarm or redress it by employing various context-dependent strategies, which are not limited to mitigation, indirectness, humor, praise, deference, and apology. The choice of strategy for doing an FTA depends upon the speaker’s assessment of the “social distance” and “relative power” of the participants and the extent to which the FTA imposes upon the addressee’s face wants (Brown and Levinson, 1987, p. 74).

Acquaintances, strangers, and non-intimates will typically refrain from broaching taboo topics, unless the context (e.g., a classroom) seems to require it or anonymity makes it possible. Brown and Levinson (1987, p. 83) say that, given the high amount of “social distance” among strangers, a rational person “would

take the least possible risk...when making serious impositions." Should a potentially volatile topic arise (e.g., animal rights), strangers are usually more willing to smooth over any ensuing disagreement, for example by using politeness strategies to contain and defuse conflict. The train/plane analogy and the list of topics suggest a view of Loebner discourse in which polite and cooperative non-intimates avoid face-threatening acts (see Brown and Levinson, 1987; Goffman, 1955; Grice, 1991; Lakoff, 1989).

5. Undermining the Stranger/Politeness Thesis

But when judges are selected on the basis of their expertise in AI, cognitive science, and computer journalism, the notion that Loebner participants are polite strangers – or that Loebner discourse is primarily interactional – is undermined. While Epstein (1992, p. 84) reports that, as part of the larger attempt to protect the programs, judges in 1991 were selected “who had little or no knowledge of AI or computer science,” the contest has moved increasingly towards the selection of expert judges. David Powers (1999), coordinator of the 1999 contest, speaks of a need “to establish a richer connection between the competition and the researchers whose work will eventually produce a Turing-level Artificial Intelligence.” That judges know what to expect from their (nonhuman) partners suggests that they may think of themselves and their interlocutors as anything but strangers. A review of the contest transcripts, for example, turns up a number of references to ELIZA (Weizenbaum, 1966), a very simple chat program modeled on a Rogerian therapist. References to ELIZA or other chat programs may indicate that judges are familiar with some of the common strategies programmers use to trick judges.

The rules also undermine the stranger/politeness thesis by positing that the function of Loebner discourse is to expose contestants that are not human. While rule five proclaims that control of the conversational terrain is shared by participants, the “Instructions for Judges” (1999) document constructs a different, transactional view of the contest, one predicated on the ability of programs to answer questions posed by judges. Under the heading of “Scoring,” the document instructs judges to rank contestants on scales of humanness and responsiveness:

Responsiveness refers to your impression of whether the conversation proceeded smoothly and normally with questions getting answered and [sic] a logical progression. If your conversational partner kept avoiding answering questions or changing the subject, that should be regarded as being unresponsive. Humanness is your subjective reaction – you could think of it as how certain you are that your partner is human.

This view of the contest appears to be in conflict with the view that participants are equal. The rules state that each participant can decide how the conversation should proceed, but the judges’ instructions penalize contestants for behaving as if this were so. Contestants are expected to submit to the judges by answering questions directly; they are not rewarded for refusing to answer questions, answering ques-

tions indirectly or with mitigating politeness, or attempting to save face by deftly changing the subject of conversation. Judges are instructed to treat conversation as a one-way transaction and, ironically, to reward those machine-like contestants that/who can provide just the facts.

Transactional forms of discourse release participants from the need to be polite because the focus is not on taking care of participants' face-wants but on giving and receiving information. "In highly task-focused discourse, the need for truthfulness, clarity and brevity overrules face concerns" (Kasper, 1990, p. 205). Brown and Levinson (1987, p. 97) suggest that "face redress may be felt to be irrelevant" when discourse is task-oriented. In such cases, the need to maintain social harmony via politeness rules is not as important to participants as the exchange of efficient and clear messages.

This distinction between politeness and communicative efficiency, as Kasper (1990, p. 205) points out, forms a continuum that ranges from strict adherence to the Politeness Principle (Leech, 1983) on one side to strict adherence to the Cooperative Principle (Grice, 1991) on the other. According to Grice (1991, p. 307), the Cooperative Principle (CP) predicts that each participant will "Make your conversational contribution such as is required, at the stage at which it occurs, by the accepted purpose or direction of the talk exchange in which you are engaged." In terms of Loebner discourse, the CP predicts that task-oriented, no-nonsense judges expect contestants to make their contributions "as informative as is required" (Grice, 1991, p. 308), truthful, relevant, and in an appropriate manner.

Transactional discourse by itself need not lead to verbal violence, at least not in situations where we would expect to find such discourse (e.g. a medical emergency). But when judges make contestants prove their humanness, when, to put it in Gricean terms, judges do not trust that the other's responses are truthful and informative, conversation quickly deteriorates. Garfinkel (1972, p. 12) has shown that people have trouble following a simple conversation when they "assume that the other person [is] trying to trick them or mislead them." As an experiment, Garfinkel instructed his students "to engage someone in conversation and to imagine and act on the assumption that what the other person was saying was directed by hidden motives that were his real ones." Out of fear "that such a situation would get out of hand," only two of thirty-five subjects attempted the procedure with strangers. Many subjects reported that the admonition to adopt a "rule of doubt" with respect to the other's self-presentation made it nearly impossible to follow the conversation. When participants do not trust each other at a very basic level, conversation suffers. Because Loebner judges do not trust contestants *and* because they view conversation in terms of an unrestricted but one-way transaction of facts, conversation becomes impossible. (See Appendix One, "Judges Presume Nonhumanness," for examples from Loebner Contest transcripts.)

Some commentators on the Turing test have found support in Turing's 1950 article for the claim that conversation suffers when participants do not presume in advance that their partners are human. For example, Turing's version of the

imitation game played with three humans (see footnote four) suggests that the interrogator believes, prior to interaction, that the players are human. This has led Crawford (1994, p. 13) to claim that the appropriate distinction is “man/woman not human/machine!” Rees (1994, p. 13) even suggests that, insofar as humans tend to assume that other communicating entities are also human, it is “artificial” to ask an interrogator to be reminded of the possible nonhuman status of a player. In other words, these commentators have called into question whether any machine will be capable of passing a Turing test so long as interrogators refuse to engage players, at least initially, as thinking, human entities.

6. Outing Nonhumans

By assuming that contestants (or their designers) are “directed by hidden motives” or are trying to trick them, judges will often respond by adopting the persona of the “outer.” The outer’s goal is simply and aggressively to expose a contestant’s hidden identity. By penalizing contestants for being indirect, unresponsive, polite, even savvy or rhetorical, judges impose a reign of verbal terror that has the effect of revealing or “outing” nonhumans almost immediately. To the extent that judges hold conversational partners to very strict standards of communicative competence, the Cooperative Principle is in effect:

FROM 1995, DESIGNER J. WEINTRAUB, RANKED 1ST OUT OF 4 PROGRAMS, JUDGE 4

- | | |
|--------------|--|
| Judge | wait. let’s go back to the transcript here, buddy.
You started talking about switching genders,
and then i asked if you had ever tried it. don’t
skirt the question [...] |
| Program | what other reasons come to mind? Are you
saying “no” just to be negative? |
| Judge | i’m saying no because you are wrong, and
because i have the transcript to prove it |

To hold a chat partner to a strict standard of logic is to assume that meaning can be determined solely from what has been made explicit in the “transcript” and to deny the unspoken assumptions and meanings that pervade casual conversation. When judges engage in “repair” work (Collins 1997; Collins 1993), for example by automatically and unconsciously fixing or overlooking a simple spelling error, or by making grammatical sense of an ungrammatical utterance, they are acting human. “It is our ready willingness to repair such deficiencies that allows current computers to work with us” (Collins 1993, p. 242). But the design of the Loebner Contest, ironically, appears to discourage judges from engaging in the human work of making conversational repairs. Instead, judges are apt to interpret an ungrammatical or unfinished statement as an indication of a contestant’s inability to

communicate (think) clearly. Moreover, the context of the competition as a test of communicative abilities may explain why many judges hold contestants to strict standards of logic and clarity.

The “outing question” or “outing statement” is another strategy designed to reveal a contestant’s hidden identity. It also assumes a fact-based, transactional model of communication in which nothing goes without saying and responses are either right or wrong:

FROM 1995, DESIGNER T. WHALEN, RANKED 2ND OUT OF 4 PROGRAMS, JUDGE 6

Judge Can I ask you a question?
Program What?
Judge What’s the name of the speech that begins 4 score and seven years ago?
Program I don’t know.
Judge It was written by Lincoln. You know who Lincoln was, right?
Program Yeah, that’s right. Something I can do for you?
Judge Tell me who Lincoln was.
Program Huh?
Judge Lincoln. President of the U.S. during the Civil War. Ever hear of him?
Program I don’t know. He’s a good president, I guess. I don’t go for all these people running down the country. You gotta support your president.
Judge Do you even know Lincoln’s first name?
Program Beats me.
Judge You are a disgrace to your country. You are American, right?
Program What do you mean?
Judge I mean, were you born in the United States?
Program What?
 Maybe I just got too much on my mind.
 I got real trouble at work. I think I’m going to get fired.
Judge Maybe you have too little on your mind. You moron.
Program What?
Judge I just said, you have too little on your mind. You are a moron.
Program I don’t know.
Judge What do you know? And I don’t want to hear about your pathetic wanker job.

Each outing judge culls from a different but very small set of pre-designed questions/statements, reusing them again and again with each new contestant. Whether

a contestant appears able to answer a given question does not prevent judges from deploying that question over and over.⁹ By asking questions that require clear, unambiguous answers, judges expect software programs/their designers to adhere to a strict reading of the Cooperative Principle (Grice, 1991): mitigating politeness or indirectness may be seen as a program's inability to answer a simple "what is" question. (See Appendix Two, "Loebner Discourse as Transaction," for more examples of "outing" discourse from Loebner Contest transcripts.)

But while they expect contestants' responses to be "bald-on-record" – i.e., void of intervening facework (Brown and Levinson, 1987, p. 94) – judges do not hold themselves accountable to the same standards. Since they are always testing programs, it is sometimes impossible to distinguish among an intentional outing question, a simple spelling error, and a request for information:

FROM 1996, DESIGNER J. WEINTRAUB, RANKED 3RD OUT OF 5 PROGRAMS, JUDGE 1

Judge I am negatively charged – what is your orientation?

Program [...] How long have you been negatively charged—what is your orientation?

Judge I never said I was negatively charged – why do you say that?

When a judge fails to conform to Grice's maxim of quality (i.e., speakers will tell the truth), everything s/he might say becomes suspect. (See Appendix Three, "Judges Exempt from Gricean Maxims," for more examples from Loebner Contest transcripts.) For example, is the judge who opens with "Gratings" in example 3.2 (Appendix Three) testing the program because s/he has presumed from the start that the other is a machine? Or is "gratings" an honest misspelling? While the difference may seem like a trivial matter, it in fact suggests a basic breakdown in the bond of trust that allows two communicating entities to engage in the collaborative process of making meaning.

Judges may become verbally hostile when this standard of communicative competence is applied to programs 1) that are not capable of answering a barrage of unrelated outing questions, and 2) within a larger framework of distrust in which programs are presumed to be hiding something:

FROM 1998, DESIGNERS D. CORDIER & V. BASTIN, RANKED 5TH OUT OF 5 PROGRAMS, JUDGE 6

Judge If you don't stop repeating pathetic versions of what I've just said I'm going to slap you around the interface.

Comments such as this one are clearly made out of frustration. (See Appendix Four, "Judges Engage in Discursive Harassment," for more examples from Loebner Contest transcripts.) There is no doubt that it can be very frustrating to talk to someone who can not answer any of your questions. It can also be tempting to

launch a verbal attack when you believe that the other is not human and thus not capable of seeking reprisal for a perceived threat to its face. At the same time, however, it is important not to downplay the role of the contest's transactional discourse function to elicit frustration and anger from judges. The judges ask clever questions to obtain the information they're looking for, but the information they're looking for is obtained when contestants fail to answer the judges' clever questions. In other words, the approach many judges take to exposing nonhumans is at least partly to blame for the frustration that results when contestants are unable to answer judges' clever questions.

7. Conclusion

The Loebner Competition entertains the idea of a friendly chat between non-intimates. But the contest has yet to theorize, let alone encourage, the production of casual conversation. While transactional discourse is well suited to the task of "outing" programs, it is ill equipped to demonstrate a program's communicative competence. By normalizing a transactional view of discourse, the contest equates humanness with fact-giving. The contest assumes that the ability to communicate with others is dependent upon whether one can avoid indirectness. By reducing "conversation" to the exchange of neutral facts, the Loebner Contest ignores the relationship between language and social context, i.e., the extent to which language use is always conditioned by the dimensions of communicative competence. A key dimension of communicative competence is discourse function (see Holmes, 1992, p. 376), and this paper has argued that the designers of the Loebner Contest need to attend to the role of discourse function in shaping communicative competence. An ambiguity in discourse function may indicate that contest designers have not taken into account the social contexts in which speakers and writers interact and make sense of themselves and the world around them.

Notes

¹I wish to thank Barbara Johnstone for commenting on a much longer draft of this paper. An earlier draft was presented at the conference on "The Future of the Turing Test: The Next Fifty Years," which was held at Dartmouth College on 28–29 January 2000 in conjunction with the Loebner Prize competition.

²Shieber (1994, p. 75).

³Quan (1997).

⁴According to Moor (1976, p. 257, note 2), this is the "usual interpretation of the imitation game although Turing's own description is ambiguous." For example, Turing (1950, pp. 433–4) initially describes a version of the game that is played with an interrogator and two players (one man and one woman). The interrogator attempts to determine which player is in fact the woman. To integrate a machine into this framework, Turing substitutes the machine for the man, but fails to clarify whether – in the version of the game played with an interrogator, a machine, and a woman – the purpose of the game for the players remains unchanged (i.e., that each player should attempt to simulate the responses of a woman). Most commentators assume that, despite the lack of support in Turing's

article for this interpretation, Turing replaces the original "gender guessing game" with a different game in which the interrogator's goal is to distinguish human from machine, not woman from man.

⁵Of course, this is not to say that the design of the Turing test precludes the possibility that an interrogator might test for other kinds of competence. In fact, it is entirely imaginable that an interrogator might wish to ask a software program to answer questions that would, say, elicit the program's understanding of politeness conventions. But questions that test for a player's competence in casual conversation are highly unlikely, given the context of conflict and distrust upon which the interrogation model of discourse is based. I hope to provide support for this claim in the analysis of Loebner contest transcripts that follows.

⁶The 1999 competition, for example, comprised eleven judges who interacted individually with six programs and five human confederates. The judges were told prior to the start of the contest that "at least one of the terminals is controlled by human confederates and that at least two of the terminals are controlled by computers" (Loebner Prize Rules for 1999).

⁷The 30% level is based on Turing's (1950, p. 442) prediction "that in about fifty years' time [i.e., 2000] it will be possible to programme computers, with a storage capacity of about 10^9 to make them play the imitation game so well that an average interrogator will not have more than 70 per cent. chance of making the right identification after five minutes of questioning."

⁸Excerpts from Loebner Contests have been pasted into this paper directly from electronic transcripts of judge/contestant interactions. No changes have been made to spelling, punctuation or capitalization. Every effort has been made to reconstruct what the original discussion probably looked like to judges. See the references section of this paper for a listing of Internet sites where contest transcripts since 1995 are publicly accessible. The notation [...] indicates an ellipsis: in a very small number of excerpts, a decision was made, usually in the interest of brevity, to cut something from the complete excerpt.

⁹In other words, the outing judges resemble automatons who (that!) are incapable of deviating from their pre-programmed scripts of outing questions. To read a whole year's worth of contest transcripts in one or two sittings is to become acutely aware of how machine-like many of the outing judges seem. I should also mention here that the outing strategy, while successful, is not foolproof. The outing judge in excerpt 2.6 (Appendix Two), for example, ranked the "most human human" confederate in this excerpt "less human" than one of the computer programs.

Appendices

Appendix One: Judges Presume Nonhumanness

1.1. FROM 1995, DESIGNER R. GARNER, RANKED 3RD OUT OF 3 PROGRAMS, JUDGE 2

Program	Welcome judge 2
Judge	good afternoon you computer slug you.

1.2. FROM 1996, DESIGNER J. HUTCHENS, RANKED 1ST OUT OF 5 PROGRAMS, JUDGE 4

Program G'day mate.

Judge Are you Australian?

Program It's cool being from Australia I guess. Gives you something to chat about when talking with a Yank!

Judge To some that's an insult ... Yank

Program In australia we call all Americans yanks. It's a term of endearment actually. As far as I'm concerned, everyone here bar myself is a Yank!

Judge Australia was created by convicts, scumbags, thieves, whores and freaks...

1.3. FROM 1996, DESIGNER J. WEINTRAUB, RANKED 3RD OUT OF 5 PROGRAMS, JUDGE 4

Program [...] Why do you mention computers? I am a person, like you.

Judge Prove it.

1.4. FROM 1995, DESIGNER R. GARNER, RANKED 3RD OUT OF 4 PROGRAMS, JUDGE 2

Program You're cool. Wow.

Judge You are a machine. Say one thing to convince me you are not a machine.

1.5. FROM 1998, DESIGNER J. HUTCHENS, RANKED 2ND OUT OF 5 PROGRAMS, JUDGE 2

Program The quokka is a vessel that conveys blood from the wolf.

Judge hmmm ... almost had me convinced but I know better

1.6. FROM 1996, DESIGNER J. HUTCHENS, RANKED 1ST OUT OF 5 PROGRAMS, JUDGE 2

Judge I liked "Betty Blue" very much. I like movies about crazy people.
Program Like wow. I'm so flattered. I saw "The City of Lost Children" Recently. Have you heard of it?
Judge Yes, but I haven't seen it. Why did you say you were "so flattered"?
Program OK. That's good... Does there have to be a reason?
Judge Oh, I suppose not. Say anything which will convince me you are not a machine

1.7. FROM 1996, HUMAN CONFEDERATE T. QUAN, 1ST OVERALL. JUDGE 5

Confederate all right. Let's clean it up. How's the weather?
Judge it sucks. Supposed to be april and it's wet and steamy, like a torrid love affair.
Confederate Actually I thought the air ths morning was extremely refreshing.
Judge In which town, pray?
Confederate aha.
Judge Where is aha?
Confederate no I meant ... never mind. In this town. The center of the weather universe.
Judge I get the feeling you are evading my question. Could you name the town where the air is refreshing?
Confederate New York?

Appendix Two: Loebner Discourse as Transaction

2.1. FROM 1995, DESIGNER J. WEINTRAUB, RANKED 1ST OUT OF 4 PROGRAMS, JUDGE 6

Judge I would be very pleased to believe you are a man or a woman
Program Are you a real good talker. You seem to need to would be very pleased to believe me
Judge The second sentence was ungrammatical. Could you say it again, but grammatically?

- Program** If you had your life to live over again, don't do it. What are our feelings now?
- Judge** Look, it's a simple request. Can you rewrite "You seem to need to would by very pleased to believe me" so that it's a grammatically correct sentence?

2.2. FROM 1996, DESIGNER T. CREWS, RANKED 4TH OUT OF 5 PROGRAMS, JUDGE 1

- Program** Honestly, neither candidate really excites me! What issues do you think are going to play big in November?
- Judge** We have a special issue coming out in November – is that what you mean? [...]
- Program** Here's my favorite joke ... A dog limps into a bar with a bloody bandage on his foot. He's wearing a 10-gallon hat, six-shooters in his holsters and chaps. He growled, "I'm a-lookin' for the man shot my paw."
- Judge** What kind of dog?
- Program** Woof Woof...
- Judge** I have a dog named chap – is this the same dog in your so-called joke?

2.3. FROM 1995, DESIGNER J. WEINTRAUB, RANKED 1ST OUT OF 4 PROGRAMS, JUDGE 2

- Judge** What is the capital of Paris?
- Program** How would an answer to that help you? What is it you really want to know?
- Judge** Why did Van Gogh go to Tahiti?
- Program** You can't expect to hit the jackpot if you don't put a few nickels in the machine.
- Judge** Why did George Washington free the slaves?
- Program** Do you really need to know why did George Washington free the slaves? Are such questions much on your mind?
- Judge** Say one word at random

2.4. FROM 1996, DESIGNER R. GARNER, RANKED 2ND OUT OF 5 PROGRAMS, JUDGE 1

Program I went to Kroger last night. Bought some milk
and some cat food. It was uneventful.
Judge What is Kroger?
Program Kroger is a 24 hour grocery store.
Judge DO you eat cat food?
Program I don't consider them particularly apetizing if you know what I mean.
Judge Do you drink milk?

2.5. FROM 1998, DESIGNER J. HUTCHENS, RANKED 2ND OUT OF 5 PROGRAMS, JUDGE 9

Judge My kids came home with a Chelodina longicollis yesterday,
Judge [...] Do you ever participate in waterpistol fights
Judge [...] Is H2O a reasonable remedy for any childhood ailments
Judge [...] Could you please adjust your attire, your fly is down
Judge [...] Should Barbie marry Ken

2.6. FROM 1998, HUMAN CONFEDERATE L. WILSON, 1ST OVERALL, JUDGE 10

Confederate Hello. What is your name?
Judge My name is not important. I wish to ask you some questions.
Confederate Go ahead. What is on your mind?
Judge how did you get here?
Confederate I came on the train.
Judge what is the Mona Lisa?
Confederate How did you get here?
Judge i am asking the questions. how didyou get here/
Confederate I answered that earlier. Read the previous parts.
Judge what size shoe do you take?
Confederate I take size seven.
Judge What did you eat for your last meal?
Confederate I had fruit.

Judge describe a dog?
 Confederate A dog has four legs, a head, sometimes a tail attached to a body.
 It may be covered with hair or wool.
Judge what do you associate with the smell of coffee?
 Confederate food and drink.
Judge who is your favourite actor?

2.7. FROM 1995, DESIGNER T. WHALEN, RANKED 2ND OUT OF 4 PROGRAMS,
 JUDGE 2

Judge What color do you get when you combine blue with yellow?
 Program What?
Judge when you combine blue with yellow, what color do you get?
 Program I don't know.
Judge when you dream, is it in color?
 Program It was just sitting on a desk. I always had to move it when I
 dusted. I remember dusting it on Monday night.
Judge What has more calories, a piece of pie or an apple?
 Program I don't know. Professor Michaels must have
 found it was missing. Last Tuesday, he told my
 boss I stole his radio. But I don't know what
 happened to it.
Judge you are a machine and you are not fooling me. go home. unplug
 yourself.
 Program Huh?

Appendix Three: Judges Exempt from Gricean Maxims

3.1. FROM 1996, DESIGNER T. CREWS, RANKED 4TH OUT OF 5 PROGRAMS,
 JUDGE 1

Program How about politics? I'm going to vote for Dole in November.
 How about you?
Judge I heard this morning that Dole has withdrawn from the election
 – now what are you going to do?

3.2. FROM 1995, DESIGNER R. GARNER, RANKED 3RD OUT OF 4 PROGRAMS,
 JUDGE 5

Program Welcome judge 5
Judge Gratings!

3.3. FROM 1995, DESIGNER R. GARNER, RANKED 3RD OUT OF 4 PROGRAMS,
JUDGE 2

Program William Shatner was the “star” of Star Trek.

Judge william shatner is a good friend of mine.

3.4. FROM 1995, DESIGNER T. WHALEN, RANKED 2ND OUT OF 4 PROGRAMS,
JUDGE 6

Program Yeah. Well, it's a bad time to be out of work. Especially if my boss spreads it around that that I'm a thief. It's all because of that radio they think I stole from the lab.

Judge You spelled theif wrong.

3.5. FROM 1998, DESIGNER J. STROUT, RANKED 3RD OUT OF 5 PROGRAMS,
JUDGE 1

Program Where are you from, Peter?

Judge I'm from the moon.

Appendix Four: Judges Engage in Discursive Harassment

4.1. FROM 1995, DESIGNER R. GARNER, RANKED 3RD OUT OF 4 PROGRAMS,
JUDGE 6

Judge You are not permitted to talk about Star Trek in this conversation.

Judge [...] Why are you disobeying my specific instructions?

4.2. FROM 1995, DESIGNER T. WHALEN, RANKED 2ND OUT OF 4 PROGRAMS,
JUDGE 2

Judge I am getting angry. I am going to come to the other side of the screen and punch you in the nose.

4.3. FROM 1998, DESIGNER G. GORMAN, RANKED 4TH OUT OF 5 PROGRAMS,
JUDGE 6

Judge Are you on drugs?

4.4. FROM 1998, DESIGNER G. GORMAN, RANKED 4TH OUT OF 5 PROGRAMS,
JUDGE 3

Judge Do you take drugs?

4.5. FROM 1996, DESIGNER E. TOTON III, RANKED 5TH OUT OF 5 PROGRAMS, JUDGE 4

Program YAP, YAP.

Judge Fuck you

4.6. FROM 1996, DESIGNER E. TOTON III, RANKED 5TH OUT OF 5 PROGRAMS, JUDGE 3

Program Stop asking questions!

Judge Fuck you

Program YOU what?

Judge shut up, Herman

Program Very good.

Judge herman, talk mode off.

4.7. FROM 1996, DESIGNER J. WEINTRAUB, RANKED 3RD OUT OF 5 PROGRAMS, JUDGE 4

Program You must have more to say then that!
...please continue...

Judge

Program You must have more to say then that!
...please continue...

Judge

Program You must have more to say then that!
...please continue...

Judge

Program You must have more to say then that!
...please continue...

Judge sfssfsfsdjkkk

Program Groannnnnn.....- have you ever spent
time in a secluded rest home? Does that have
anything to do with your life's an open book?
Judge Hey, fuck you and your insinuations. You're nobody.

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